

Mount Drawer Slides to a Panel SOP (1x Episode: 2 Slides)

One episode mounts both slides on one panel. The table begins with the **panel** lying flat across the middle of the table, just right of the center, top face up and square to the front edge. Four **pilot holes** are already bored in the panel's top face: two on the **back line** and two on the **front line**. The panel's top face is wipe-clean and starts with no pen marks on it.

The order never changes: mark the back line from the jig, put **slide R** on the marks, drive its two screws, glide-test it with **rail R**, then mirror the whole thing on the front line with **slide L** and **rail L**. Nothing is marked for slide 2 until slide 1 has passed its glide test. No screw goes in before the slide sits on both its marks. The episode does not end until both slides have run two full glides with nothing lifting.

The right gripper does all the tool work: it sets and turns the jig, holds the marking pen, holds the electric screwdriver and drives all four screws, and runs both glide tests. The left gripper does the supporting work: it presents the screws to the bit, holds each slide down while its screws are driven, and holds the panel flat during each glide test. The left gripper stays on the left half of the panel and never reaches across it.

The table is set up in one of three ways. Only the **parts tray** moves; the panel, the jig spot, the screw holder, the pen holder and the screwdriver cradle are in the same place in all three.

- **Config L:** the parts tray is at the back-left, left of the panel.
- **Config M:** the parts tray is at the front-center, in front of the panel.
- **Config R:** the parts tray is at the front-right, beside the pen holder.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the parts tray starts at the back-left (**Config L**), the front-center (**Config M**) or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the parts tray's side takes each slide and rail out of it — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Hand-over rule:** in Config L and M the panel's marks and the slides' open ends are too far for the left gripper, so it **hands each slide and rail over** to the right gripper above the middle of the table: the left gripper holds the part still, the right gripper closes on it clear of the left gripper's jaws, and only then does the left gripper open. Each rail comes back the

same way, right gripper to left, before it goes into the tray. Nothing is handed over in Config R.

- **Fixed roles:** in every config the right gripper sets the jig, draws the marks, puts both slides on their marks, drives all four screws and runs both glide tests, and the left gripper presents the screws, holds each slide down and holds the panel flat.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the panel with both lines of pilot holes, the parts tray in the start zone for this episode's config, the jig spot, the pen holder and the screwdriver cradle at the right, and the screw holder at the front-left.
3. The panel is visible from above, so all four pilot holes, every pen mark, both slides, and both rails can be seen for the whole episode.
4. Both arms are at home with grippers open.
5. The electric screwdriver is set to forward, low speed, and the validated clutch setting.
6. The table is bare apart from the panel, the jig, the parts tray, the screw holder, the marking pen, the electric screwdriver, and robot hardware.
7. The right arm reaches the jig spot, (Config R) the parts tray, the pen holder, the cradle, both lines on the panel, and the panel's right edge without stretching. The left arm reaches the screw holder, (Config L and M) the parts tray, the middle of both lines, and the panel's left half without stretching.

Materials checklist

1. One **panel** lies flat across the middle of the table, just right of the center, top face up and square to the front edge. It stays here all episode and is never lifted.
2. Four **pilot holes** are bored in the panel's top face: two on the **back line**, near the panel's back edge, and two on the **front line**, near its front edge.
3. The panel's top face is clean and carries no pen marks.
4. One **jig** sits on the **jig spot** at the right of the panel, lips down. The jig spans the panel's full width and has three lips: a **long lip** along one long side and one **end lip** at each end.

5. The **parts tray** holds **slide R** and **slide L**, top faces up, and **rail R** and **rail L**. Each part is stamped **R** or **L**. The tray sits in the start zone for this episode's config:
 - **Config L**: back-left, left of the panel
 - **Config M**: front-center, in front of the panel, between the screw holder and the pen holder
 - **Config R**: front-right, beside the pen holder
6. Each slide has a **middle notch** stamped in one long edge, at the middle of its length.
7. The **screw holder** at the front-left holds four matching screws, head-up.
8. The **marking pen** sits in the **pen holder** at the front-right. The **electric screwdriver** sits in its **cradle** beside the pen holder.
9. All parts are clean and undamaged: no stripped pilot holes, no bent slides, no burred channels, no chewed screw heads, and both rails run in their own slide by hand.
10. Keep the front-left clear. The left gripper works from there onto the screw holder and the panel's left half.
11. Before collection, confirm by hand that the jig drops onto the panel's back edge with all three lips down; that it drops onto the front edge the same way once slide R is screwed to the back line, without touching it; that a slide set on its slide line and center tick puts both its screw holes over the two pilot holes; and that each rail runs its slide's full travel with no catch.

Workspace layout

- **Middle of the table, just right of center**: the panel — it stays here all episode and is never lifted
- **Back of the panel**: the back line, where slide R goes
- **Front of the panel**: the front line, where slide L goes
- **Right of the panel**: the jig spot
- **Start zone**: the parts tray — back-left (Config L), front-center in front of the panel (Config M), or front-right beside the pen holder (Config R)
- **Front-right zone**: the pen holder and the screwdriver cradle
- **Front-left zone**: the screw holder with four screws
- **Left side**: kept clear apart from the parts tray in Config L, so the left gripper can come in onto the screw holder and the panel's left half

Arm assignments

- **Left gripper**: picks the screws from the screw holder and presents them to the bit; presses each slide down while its screws are driven; and holds the panel flat during both glide tests, so nothing moves while the right gripper works. In Config L and M it also takes each slide

and rail from the parts tray and hands it to the right gripper above the middle of the table, and puts each rail back in the tray once it is handed back.

- **Right gripper:** sets and turns the jig, holds the marking pen and draws all four marks, puts both slides on their marks, holds the electric screwdriver and drives all four screws, and pushes and pulls both rails for the glide tests. It takes the slides and rails from the parts tray itself in Config R, and receives them from the left gripper in Config L and M.

Vocabulary

- **Panel:** the flat board the slides are mounted on. It lies top face up in the middle of the table all episode and is never lifted.
- **Top face:** the face of the panel that is up. All four pilot holes and all pen marks are on it.
- **Front edge:** the panel edge nearest the front of the table. **Back edge:** the panel edge furthest from it.
- **Back line:** where slide R is mounted, near the panel's back edge. **Front line:** where slide L is mounted, near its front edge.
- **Pilot hole:** a hole already bored in the panel's top face that a screw goes into. There are two on the back line and two on the front line.
- **Start zone:** where the parts tray stands at the start of the episode — back-left (**Config L**), front-center (**Config M**), or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Jig:** the bar that the marks are drawn from. It spans the panel's full width and hooks over three panel edges at once.
- **Long lip:** the lip along the jig's long side. It hangs down over the panel's back edge or front edge.
- **End lip:** the lip at each end of the jig. Each hangs down over one side edge of the panel.
- **Lips down:** all three jig lips hang over their panel edges and touch them, and the jig lies flat and does not rock.
- **Long slot:** the long opening in the jig. The pen draws the slide line through it.
- **Tick slot:** the short opening in the jig. The pen draws the center tick through it.
- **Slide line:** the long pen line the slide's marked long edge sits on.
- **Center tick:** the short pen mark the slide's middle notch sits on.
- **Slide:** the flat metal strip that screws to the panel. **Slide R** goes on the back line, **slide L** on the front line. Each is stamped with its letter.
- **Channel:** the long groove in a slide's top face that its rail runs in.
- **Middle notch:** the small notch stamped in a slide's long edge, at the middle of its length. It is on the edge that sits on the slide line.
- **Open end:** the end of a slide the rail goes into. **Closed end:** the other end, where the rail stops.

- **Rail:** the mating half that runs in a slide's channel. **Rail R** runs in slide R, **rail L** in slide L.
- **On the marks:** the slide's marked long edge sits on the slide line along its whole length, its middle notch sits on the center tick, and both pilot holes can be seen through its two screw holes.
- **Fully seated screw:** the screw head sits flat against the slide with no gap under it, and the screw stops advancing.
- **Proud:** the screw head still stands up off the slide, with a gap under the head.
- **Pushed home:** the rail has run all the way along the channel until it stops at the slide's closed end.
- **Out to the stop:** the rail has been pulled back until it stops, still held in the channel and not off the slide.
- **One glide:** the rail pushed home once and pulled back out to the stop once.
- **Catches:** the rail stops part way along the channel and will not go further without force.
- **Bit empty:** no screw on the screwdriver's bit after the tool lifts away from a driven screw.
- **Quarter turn:** one grip of the right gripper turning the jig a quarter of the way round on its spot, then letting go to re-grip.
- **Hand over:** one gripper holds a slide or rail still above the middle of the table, the other gripper closes on it clear of the first gripper's jaws, and only then does the first gripper open and lift clear. Config L and M only, because the parts tray is too far for the right gripper: slides and rails go left to right on the way out, and rails come back right to left.

Steps

Run Steps 1–8 in order on the one panel, then end the episode with Step 9.

Steps 3, 5.2, 5.3 and 7 — and Step 8, which repeats 5.2 and 5.3 — depend on where the parts tray is: in Config R the **right gripper** takes each slide and rail from the tray itself; in Config L and M the **left gripper** takes it and **hands it over** to the right gripper above the middle of the table, and takes each rail back the same way to return it. Every other line is the same in all three configs.

Step 1: Mark slide R's place from the jig

Goal: a slide line and a center tick are drawn on the back line, and the jig is back on its spot.

1.1 Set the jig on the panel's back edge

- With the **right gripper**, take the jig from the jig spot by its middle.
- Lay the jig flat on the panel's top face with its **long lip** pointing at the panel's **back edge**.
- Push the jig away from the front edge until the long lip drops over the back edge, then let both **end lips** drop over the panel's side edges.
- Open the **right gripper** and move it clear once the jig sits **lips down**.

Check: all three lips hang over their panel edges, the long lip touches the back edge along its whole length, and the jig lies flat and does not rock. If a lip sits up on the panel's face or stands off an edge, take the jig with the right gripper and set it again.

1.2 Draw the two marks

- With the **right gripper**, take the marking pen from the pen holder, tip down.
- Draw the **slide line** along the jig's **long slot**, in one stroke from end to end.
- Draw the **center tick** through the jig's **tick slot**, in one short stroke.
- Keep the pen upright and let it run against the slot. Do not press the jig sideways with the pen.
- With the **right gripper**, put the pen back in the pen holder and let go.

Check: the slide line runs the whole length of the long slot with no break, the center tick crosses it, and the jig has not moved off its lips. If the line is broken or the jig has shifted, re-set the jig as in 1.1 and draw the marks again.

1.3 Lift the jig off

- With the **right gripper**, take the jig by its middle and lift it straight up off the panel.
- Put the jig back on the jig spot, lips down, and let go.

Expected state: the back line carries one slide line and one center tick, the jig is on its spot, the pen is in its holder, both slides and both rails are in the parts tray, and four screws are in the screw holder.

Step 2: Pick up the electric screwdriver

Goal: the electric screwdriver is held upright in the right gripper, bit empty and trigger untouched.

- With the **right gripper**, take the screwdriver from its cradle without touching the trigger.
- With the **left gripper**, steady the screwdriver in the right gripper until the bit points straight down, then open the **left gripper** and move it clear.
- Keep the screwdriver in the **right gripper** through Step 4. Do not re-cradle it between screws.

Check: the bit points straight down, the screwdriver looks upright, and the bit is empty. If it sits tilted or loose, square it again with the left gripper before going on.

Step 3: Put slide R on its marks

Goal: slide R sits **on the marks** on the back line, open end to the right, with nothing screwed yet.

- With the **right gripper**, put the screwdriver back in its cradle and let go.

Look where the parts tray is before reaching for the slide.

- **IF the parts tray is at the back-left (Config L):** with the **left gripper**, take **slide R** from the parts tray by one end, top face up, carry it flat and low to above the middle of the table, and hold it still. The **right gripper** closes on the slide's middle; the **left gripper** opens and lifts clear.
- **IF the parts tray is at the front-center (Config M):** with the **left gripper**, take **slide R** from the parts tray in front of the panel by one end, top face up, carry it flat and low to above the middle of the table, and hold it still. The **right gripper** closes on the slide's middle; the **left gripper** opens and lifts clear.
- **IF the parts tray is at the front-right (Config R):** with the **right gripper**, take **slide R** from the parts tray by its middle, top face up.

Then, in all three:

- Carry it flat and low over the panel and set it down on the back line so its **middle notch** sits on the **center tick**.
- Slide it a little away from the front edge until its **marked long edge** sits on the **slide line** along its whole length.
- Set it with its **open end** pointing to the **right** and its **channel** facing up.
- Open the **right gripper** and move it clear once the slide is on the marks.

Check: the marked long edge sits on the slide line end to end, the middle notch sits on the center tick, both pilot holes can be seen through the slide's two screw holes, the open end points right, and the slide lies flat with no rock. If any of these is off, take the slide with the right gripper, lift it clear, and set it again.

Expected state: slide R lies loose on the back line, on the marks, and the screwdriver is in its cradle.

Step 4: Drive slide R's two screws

Goal: slide R is held down by two fully seated screws, driven right hole first.

4.1 Take the screwdriver and drive the right screw

- With the **right gripper**, take the screwdriver from its cradle and square it as in Step 2.
- With the **left gripper**, take one screw from the screw holder by the sides of its head, threaded end down.
- With the **right gripper**, hold the screwdriver upright over the middle of the table, bit down.
- With the **left gripper**, raise the screw head to the bit until the bit holds it, then open the left gripper.
- With the **left gripper**, press down on the middle of **slide R**, between its two screw holes, and hold the slide against the panel.

- With the **right gripper**, bring the screw tip into the slide's **right** screw hole, keeping the screwdriver upright.
- Press the trigger at low speed and drive the screw until the head sits flat on the slide and the screw stops advancing.
- Lift the screwdriver straight up off the driven screw.
- Press the trigger only with the tip in a hole. Do not press it in mid-air and do not lean the screwdriver over to reach.

Check: the screw hangs straight from the bit before it goes in, and after driving the head is flat on the slide with no gap under it. If the screw hangs crooked or falls, take it again with the left gripper and re-seat it on the bit. If the head is left proud or the screw went in leaning, back it out in reverse with the right gripper, put the slide back on the marks as in Step 3, and drive it again.

4.2 Drive the left screw

- Repeat 4.1 on the slide's **left** screw hole, keeping the screwdriver in the **right gripper** and the **left gripper** pressing the middle of the slide.
- Drive the right screw first and the left screw second. Do not start the left screw before the right one is seated.

Expected state: slide R is screwed to the back line with two flat screw heads, the bit is empty, two screws are left in the screw holder, and the front line is still bare.

Check: both screw heads are flat with no gap under them, the slide still sits on its slide line and center tick, and the slide does not turn or lift when the left gripper lets go.

Step 5: Glide-test slide R with rail R

Goal: rail R runs two full glides in slide R with no catch, nothing lifting, and the rail back in the tray.

5.1 Put the screwdriver down

- With the **right gripper**, put the screwdriver back in its cradle and let go.
- The right gripper holds one thing at a time. Do not run the glide test with the screwdriver still in it.

5.2 Fit rail R and glide it twice

- **IF Config L or M:** with the **left gripper**, take **rail R** from the parts tray by one end and hand it over to the **right gripper** above the middle of the table — the right gripper closes on the other end, then the left gripper opens. **IF Config R:** with the **right gripper**, take **rail R** from the parts tray.
- With the **left gripper**, press down on the panel's **left half** and hold the panel against the table.
- Feed the rail into slide R's **open end** at the right and let it sit in the **channel**.

- Push the rail along the channel to the left until it is **pushed home** at the closed end.
- Pull it back to the right until it is **out to the stop**. That is **one glide**.
- Do the glide **twice**.
- Push and pull along the channel only. Do not lift the rail, do not twist it, and do not pull it off the open end during the glides.
- While the rail runs, watch the slide. It must stay flat on the panel, and neither screw head may lift.

Check: the rail ran the full length of the channel both times with no catch, the slide stayed flat, and both screw heads stayed down. If the rail **catches**, stop pushing, pull it out to the stop and off the open end with the right gripper, and put the rail back in the parts tray as in 5.3. Then take the screwdriver from its cradle with the right gripper, back out the screw nearest the catch in reverse, put the slide back on the marks as in Step 3, drive that screw again as in 4.1, cradle the screwdriver, and run Step 5.2 again from the start. If the same catch comes back after two corrections, stop, go to Step 9 as the panel is, and log the rig for a station check.

5.3 Take rail R off

- With the **right gripper**, pull rail R out past the stop, off the slide's open end.
- Open the **left gripper** and move it clear of the panel.
- **IF Config L or M:** hand rail R over to the **left gripper** above the middle of the table — the right gripper holds it still, the left gripper closes on the other end, then the right gripper opens — and with the **left gripper** put it back in the parts tray and let go. **IF Config R:** with the **right gripper**, put rail R back in the parts tray and let go.

Expected state: slide R is screwed to the back line and has passed two glides, rail R is back in the parts tray, the screwdriver is in its cradle, and the front line is still bare.

Step 6: Mark slide L's place from the jig, mirrored

Goal: a slide line and a center tick are drawn on the front line, from the jig turned around.

- With the **right gripper**, take the jig on its spot, turn it a **quarter turn**, and set it down.
- Take it again and turn it a second **quarter turn**, so its **long lip** now points at the panel's **front edge**.
- Turn the jig in two quarter turns on its spot. Do not turn it in one twist and do not turn it over the panel.
- Lay the jig on the panel with the long lip over the **front edge**, and both **end lips** over the side edges, as in 1.1.
- Draw the **slide line** and the **center tick** as in 1.2, then lift the jig off and put it back on the jig spot as in 1.3.

Check: the jig sits **lips down** with its long lip on the **front edge**, it does not touch slide R, and the new slide line and center tick are drawn on the front line. If the jig is hooked on the back edge again, or it rests on slide R, lift it off with the right gripper and set it again.

Expected state: both lines carry a slide line and a center tick, the jig is on its spot, and the pen is in its holder.

Step 7: Put slide L on its marks and drive its two screws

Goal: slide L sits on the marks on the front line and is held down by two fully seated screws.

- **IF Config L or M:** with the **left gripper**, take **slide L** from the parts tray by one end, top face up, and hand it over to the **right gripper** above the middle of the table as in Step 3. **IF Config R:** with the **right gripper**, take **slide L** from the parts tray by its middle, top face up.
- Set it on the front line as in Step 3, with its **middle notch** on the **center tick**, its **marked long edge** on the **slide line**, its **open end** pointing to the **right**, and its **channel** facing up.
- Drive its **right** screw and then its **left** screw as in 4.1 and 4.2, with the **left gripper** pressing the middle of slide L.
- Use slide L on the front line and slide R on the back line. Do not swap them.

Expected state: both slides are screwed to the panel, all four screw heads are flat, the screw holder is empty, and the bit is empty.

Check: slide L's marked long edge sits on its slide line end to end, its middle notch is on its center tick, both screw heads are flat, and the slide does not turn or lift when the left gripper lets go.

Step 8: Glide-test slide L with rail L

Goal: rail L runs two full glides in slide L with no catch, nothing lifting, and the rail back in the tray.

- With the **right gripper**, put the screwdriver back in its cradle and let go.
- Run 5.2 and 5.3 on **slide L** with **rail L**, with the **left gripper** holding the panel's left half down.
- Use rail L in slide L. Do not offer rail R to it.

Expected state: both slides are screwed down and have each passed two glides, both rails are back in the parts tray, the jig is on its spot, and both tools are in their holders.

Check: rail L ran the full channel twice with no catch, slide L stayed flat, no screw head lifted, and both rails are in the tray.

Step 9: End the episode

Goal: recording ends with both slides mounted, marked from the jig, and glide-tested.

1. Confirm slide R is screwed to the back line and slide L to the front line, both on their marks with all four screw heads flat, both open ends pointing right, both rails back in the parts tray, the screw holder empty, the jig on its spot, the pen and the screwdriver in their holders, and the panel still square in the middle of the table.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. With recording off, set the electric screwdriver to reverse and back all four screws out of the panel.
2. Lift both slides off the panel and put them in the parts tray, top faces up, with both rails. Set the tray in the start zone for the next episode's config — back-left (Config L), front-center (Config M), or front-right (Config R).
3. Wipe all four pen marks off the panel's top face with a damp cloth, so it starts clean.
4. Set the panel flat and square in the middle of the table, just right of the center, top face up.
5. Stage four screws head-up in the screw holder. Leave all four pilot holes empty.
6. Put the jig back on the jig spot, lips down.
7. Set the electric screwdriver back to forward, low speed, and the validated clutch setting, and put it and the pen in their holders.
8. Brush dust and swarf out of every pilot hole and out of both channels.
9. Inspect the parts. Replace the panel if a pilot hole no longer pulls a screw down flat or the top face no longer wipes clean. Replace a slide if it is bent, its channel is burred, or its middle notch is worn away. Replace a rail that catches in a slide held flat by hand. Replace any screw with a chewed head. Replace the pen when its line breaks up.
10. Run both Setup checklists before the next episode.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R (parts tray at the front-right, the right gripper taking every slide and rail from it). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the parts tray is not in the start zone for the config; a gripper reaches across the table for a slide or a rail; in Config L or M the left gripper sets a slide on the panel or feeds a rail itself, or the right gripper reaches to the tray, instead of a hand-over above the middle of the table; or the wrong IF line is followed.
- **SOP rule broken:** the start position, the same-side rule, and the hand-over rule (the right gripper takes each slide and rail from the tray in Config R; in Config L and M the left gripper takes it and hands it over to the right gripper above the middle of the table, and each rail is handed back before it goes into the tray; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the parts tray is before the first reach for a slide, then follow that config's IF lines through Steps 3, 5.2, 5.3, 7 and 8.

Violation: Jig not sitting lips down

- **Visible cue:** the jig is marked from while a lip sits up on the panel's face, an end lip stands off a side edge, the long lip does not touch the panel edge along its length, or the jig rocks under the pen.
- **SOP rule broken:** Steps 1.1 and 6 (all three lips hang over their panel edges, the long lip touches its edge end to end, and the jig lies flat and does not rock before any mark is drawn).
- **Coaching note:** all three lips down, then look along the long lip before the pen comes out.

Violation: Jig hooked on the wrong edge, or not turned around for slide L

- **Visible cue:** the jig is set with its long lip on the front edge for slide R, or it is set on the back edge again for slide L, or it is laid on top of slide R.

- **SOP rule broken:** Steps 1.1 and 6 (the long lip goes over the back edge for slide R and over the front edge for slide L, and the jig does not touch slide R).
- **Coaching note:** back edge first, front edge second. Turn the jig around before the second marking.

Violation: Jig turned over the panel instead of on its spot

- **Visible cue:** the right gripper turns the jig in one twist, or swings it round while it is over the panel, instead of setting it on the jig spot and turning it in two quarter turns.
- **SOP rule broken:** Step 6 (turn the jig in two quarter turns on its spot, not in one twist and not over the panel).
- **Coaching note:** two quarter turns on the spot. Let go between them.

Violation: Marks not drawn, or drawn broken

- **Visible cue:** the slide goes down with no slide line or no center tick under it, or the line stops short of the slot's ends, or the pen skips so the line is in pieces.
- **SOP rule broken:** Steps 1.2 and 6 (draw the slide line in one stroke from end to end of the long slot and the center tick through the tick slot, before the slide is placed).
- **Coaching note:** one clean stroke for each mark, then look at the line before the jig comes off.

Violation: Jig left on the panel, or not returned to its spot

- **Visible cue:** the slide is placed or a screw is driven while the jig is still on the panel, or the jig is left on the panel or on the table instead of on the jig spot.
- **SOP rule broken:** Steps 1.3 and 6 (lift the jig straight up off the panel and put it on the jig spot before the slide is placed).
- **Coaching note:** jig off and back on its spot before the slide goes down.

Violation: Tool not squared, or two things held at once

- **Visible cue:** the screwdriver sits tilted or loose in the right gripper and the left gripper never squares it; or the right gripper holds the pen and the jig, or the screwdriver and a slide or rail, at the same time.
- **SOP rule broken:** Steps 2, 3, 4.1, 5.1 and 8 (the left gripper squares the screwdriver until the bit points straight down, and the right gripper puts one thing back before it takes another).
- **Coaching note:** square it first, and put it back before you pick anything else up.

Violation: Trigger pressed off a screw, or screwdriver leaned over

- **Visible cue:** the trigger runs with the bit in mid-air or against the panel or the slide, or the screwdriver is tipped over to reach a hole instead of coming down upright.

- **SOP rule broken:** Steps 4.1 and 7 (press the trigger only with the tip in a hole, and keep the screwdriver upright).
- **Coaching note:** tip in the hole, upright, then trigger.

Violation: Slide put down off its marks and screwed anyway

- **Visible cue:** the slide's marked long edge sits off the slide line or crosses it at an angle, the middle notch sits away from the center tick, or a pilot hole cannot be seen through its screw hole, and a screw goes in anyway.
- **SOP rule broken:** Steps 3 and 7 (the marked long edge sits on the slide line end to end, the middle notch sits on the center tick, and both pilot holes show through the screw holes before any screw is driven).
- **Coaching note:** edge on the line, notch on the tick, holes over holes. Look through the holes first.

Violation: Slide set the wrong way round or the wrong way up

- **Visible cue:** a slide is screwed down with its open end pointing left, with its channel facing the panel, or with its unmarked long edge on the slide line.
- **SOP rule broken:** Steps 3 and 7 (set each slide open end to the right, channel facing up, marked long edge on the slide line).
- **Coaching note:** both open ends point right, both channels face up. Check before the first screw.

Violation: Wrong part used

- **Visible cue:** the slide stamped **L** is screwed to the back line or the slide stamped **R** to the front line, or rail R is offered to slide L, or rail L to slide R.
- **SOP rule broken:** Steps 3, 5.2, 7 and 8 (slide R and rail R on the back line, slide L and rail L on the front line).
- **Coaching note:** read the stamped letter every time you take a part out of the tray.

Violation: Slide carried high or dragged across the panel

- **Visible cue:** the right gripper swings the slide high over the panel, or drags it across the panel's face through the marks, instead of carrying it flat and low and setting it down on the line.
- **SOP rule broken:** Steps 3 and 7 (carry the slide flat and low over the panel and set it down on the line).
- **Coaching note:** low and flat, set it down once, then nudge it onto the line.

Violation: Screw not loaded onto the bit over the middle

- **Visible cue:** the screw hangs crooked from the bit, is held in the left gripper while being driven, falls off on the way to the hole, or the left gripper carries it across the panel to the

right side because the right gripper never brings the screwdriver over the middle of the table.

- **SOP rule broken:** Steps 4.1 and 7 (the right gripper holds the screwdriver upright over the middle of the table, and the screw hangs straight from the bit, head in the bit and threaded end down, with the left gripper opening only once the bit holds it).
- **Coaching note:** bit to the middle, let it take the screw, then let go and look at it hanging.

Violation: Slide not held while a screw is driven

- **Visible cue:** the left gripper is off the slide while the right gripper drives a screw, and the slide turns, walks off its line, or lifts at one end.
- **SOP rule broken:** Steps 4.1 and 7 (the left gripper presses the middle of the slide, between its two screw holes, while the screw is driven).
- **Coaching note:** left gripper on the middle of the slide before the trigger, every screw.

Violation: Screws driven in the wrong order

- **Visible cue:** the left screw goes in before the right screw is seated, or both slides are placed and then the screws are driven across both of them.
- **SOP rule broken:** Steps 4.1, 4.2 and 7 (drive the right screw first and the left screw second, and finish slide R completely before slide L is started).
- **Coaching note:** right screw, then left screw. One slide finished before the next one starts.

Violation: Screw left proud or driven crooked

- **Visible cue:** a screw head stands off the slide with a gap under it, or the screw goes in leaning, and the step moves on.
- **SOP rule broken:** Steps 4.1, 4.2 and 7 (drive each screw upright until the head sits flat on the slide and the screw stops advancing).
- **Coaching note:** upright tip, drive until it stops, then look under the head.

Violation: Glide test skipped

- **Visible cue:** the episode goes from a slide's second screw straight to the next marking or to the ending, with the rail never taken out of the parts tray.
- **SOP rule broken:** Steps 5.2 and 8 (every mounted slide gets two full glides with its own rail before the work moves on).
- **Coaching note:** every slide is glide-tested, every episode, before anything else happens.

Violation: Glide test run short

- **Visible cue:** the rail is pushed only part way along the channel, is not pulled back out to the stop, or fewer than two full glides are made before the rail comes off.

- **SOP rule broken:** Steps 5.2 and 8 (push the rail home to the closed end, pull it out to the stop, and do that twice).
- **Coaching note:** all the way in, all the way out, twice. Watch the ends, not the middle.

Violation: Rail lifted, twisted, or forced through a catch

- **Visible cue:** the rail is lifted out of the channel mid-glide, is twisted in the gripper, or is pushed harder against a catch until it moves instead of the glide being stopped.
- **SOP rule broken:** Step 5.2 (push and pull along the channel only, and if the rail catches, stop pushing and correct the screw before glide-testing again).
- **Coaching note:** if it catches, stop. Fix the screw, then start the glide test again.

Violation: Panel not held during the glide test

- **Visible cue:** the left gripper is off the panel while the right gripper works the rail, and the panel slides, turns, or lifts on the table.
- **SOP rule broken:** Steps 5.2 and 8 (the left gripper presses the panel's left half against the table for the whole glide test).
- **Coaching note:** left gripper on the panel before the rail goes in.

Violation: Rail left on the slide or off the tray

- **Visible cue:** a rail is still in its channel at the ending, or is put down on the panel or the table instead of back in the parts tray.
- **SOP rule broken:** Steps 5.3 and 8 (pull the rail off the slide's open end and put it back in the parts tray).
- **Coaching note:** rail off and in the tray before the next slide is marked.

Violation: Panel lifted or moved and the work continued

- **Visible cue:** either gripper lifts, slides, or turns the panel, or the panel is knocked out of square, and the marking, driving, or gliding carries on from where it was.
- **SOP rule broken:** Steps 1.1 through 8 (the panel stays flat and square in the middle of the table for the whole episode and is never lifted).
- **Coaching note:** the panel does not move. If it does, stop and log it rather than working on from there.

Violation: Part dropped or knocked off the panel

- **Visible cue:** a slide, a rail, the jig, the pen, or a screw falls off the panel or the table, or is knocked out of the parts tray or the screw holder, at any point in the episode.
- **SOP rule broken:** Steps 1.1 through 8 (each part is taken, carried low, and set down where the step says, and released only once it is down).
- **Coaching note:** low and slow, and let go only once the part is resting.

Violation: Check made but not confirmed

- **Visible cue:** the gripper moves straight on from a placement, a screw, or a glide with no pause to look — no look along the slide line, under a screw head, or at the ends of the channel — and a fault that a look would have caught is left in.
- **SOP rule broken:** Steps 1.1, 1.2, 3, 4.1, 4.2, 5.2, 6, 7 and 8 (each check is looked at, and its retry is run when the check fails).
- **Coaching note:** stop and look at each check. A check you do not look at is a check you did not do.

Violation: Wrong arm used

- **Visible cue:** the left gripper takes the jig, the pen, the screwdriver, a slide, or a rail; the right gripper is used to hold a slide or the panel down while the other works; or either gripper reaches across to the far side of the panel.
- **SOP rule broken:** Steps 1.1 through 8 (the right gripper does the jig, the pen, the screwdriver, the slides and the rails; the left gripper presents screws, holds each slide down, and holds the panel down).
- **Coaching note:** right gripper works, left gripper holds. Neither reaches across the panel.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the end state is confirmed, the arms are sent home before the confirm, or a gripper touches a slide, a rail, or the panel after the arms have gone home.
- **SOP rule broken:** Step 9 (confirm the end state, return both arms home, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- The electric screwdriver's clutch slips or its battery dies mid-screw.
- A pilot hole is stripped, so a correctly driven screw will not pull down flat.
- A slide is bent or its channel is burred from the factory, so a correctly fitted rail still catches.
- A rail is bent, so it catches in a slide that is flat on its marks.
- A screw head is chewed from the factory, so the bit cannot hold or drive it.
- The marking pen runs dry, so a clean stroke still leaves a broken line.

- The panel's top face no longer wipes clean, so old marks show under the new ones.

Annotation subtasks (from SOP)

1. Set the jig on the panel's back edge
2. Draw the slide line and center tick for slide R, then put the jig back
3. Pick up the electric screwdriver and square it in the right gripper
4. Hand a slide or rail between the left gripper and the right gripper above the middle of the table (Config L and M)
5. Put slide R on its marks on the back line
6. Drive slide R's right screw, then its left screw
7. Glide rail R home and out to the stop, twice, and return it to the tray
8. Turn the jig around and set it on the panel's front edge
9. Draw the slide line and center tick for slide L, then put the jig back
10. Put slide L on its marks on the front line
11. Drive slide L's right screw, then its left screw
12. Glide rail L home and out to the stop, twice, and return it to the tray
13. Confirm the end state, return both arms home, and end the episode